

Inflation Persistence and Monetary Policy Effectiveness: Evidence from Emerging Economies

Working Paper

JEL Classification: E31, E52, E58, F41

Keywords: Inflation persistence, monetary policy, emerging economies, panel data, Taylor Rule

Abstract

This paper investigates the relationship between inflation persistence and monetary policy effectiveness in emerging economies. Using panel data from a cross-section of emerging markets, we estimate an autoregressive model of inflation and a fixed-effects panel specification that incorporates monetary policy instruments, exchange rate dynamics, and institutional factors. Our results indicate that inflation persistence is high, with the lagged inflation coefficient estimated at approximately 0.72. While contractionary monetary policy exerts a statistically significant downward pressure on inflation, its effectiveness is constrained by structural vulnerabilities including exchange rate pass-through, fiscal imbalances, and weak institutional frameworks. These findings underscore the need for complementary structural and institutional reforms alongside conventional monetary tightening to achieve durable price stability in emerging market contexts.

1. Introduction

Price stability remains one of the central objectives of macroeconomic policy across both advanced and emerging economies. Over the past three decades, a broad consensus has emerged around the desirability of low and stable inflation, reflected in the widespread adoption of inflation targeting frameworks and the strengthening of central bank independence. Despite these institutional developments, many emerging economies continue to experience elevated and persistent inflation, suggesting that structural and institutional impediments remain significant barriers to the effectiveness of monetary policy.

A key concept in this context is inflation persistence — the tendency of inflation to deviate from its long-run equilibrium for extended periods following a shock. High persistence implies that disinflation is both costly and prolonged, requiring sustained policy tightening that may generate output losses and financial instability. Understanding the determinants of inflation persistence, and the extent to which monetary policy can moderate it, is therefore of considerable policy relevance.

Emerging economies present a particularly challenging environment for monetary policy. Exchange rate volatility, fiscal dominance, limited central bank credibility, and fragile inflation expectations all interact to weaken the standard transmission channels through which policy

interest rates affect aggregate demand and inflation. As a result, the empirical relationship between monetary policy instruments and inflation outcomes in these settings remains an important and unresolved question in the literature.

This study contributes to that literature by examining inflation persistence and monetary policy effectiveness jointly within a unified panel data framework. Using annual data from a sample of emerging economies sourced from the World Bank and the International Monetary Fund (IMF), we estimate the degree of inflation persistence and assess the role of policy interest rates, exchange rate dynamics, the output gap, and fiscal variables in shaping inflation outcomes. The remainder of this paper proceeds as follows. Section 2 reviews the relevant literature. Section 3 presents the theoretical framework. Section 4 describes the data and empirical methodology. Section 5 discusses the results, and Section 6 concludes.

2. Literature Review

The measurement and determinants of inflation persistence have attracted substantial attention in the macroeconomic literature. Early contributions relied on univariate autoregressive specifications applied to consumer price index (CPI) and producer price index (PPI) data to characterize the speed of mean reversion following inflationary shocks. These studies established that persistence varies considerably across countries and time periods, and that structural breaks — often associated with changes in monetary policy regime — can significantly alter estimated persistence coefficients.

The theoretical literature emphasizes the role of monetary policy credibility and institutional design in shaping inflation dynamics. Independent and transparent central banks are better positioned to anchor expectations, thereby reducing the propagation of inflationary shocks through the wage- and price-setting process. This insight underpins the inflation targeting literature, which documents significant reductions in both inflation levels and persistence following the adoption of formal targets in a range of economies.

Emerging economies, however, present distinctive challenges that complicate the application of lessons from advanced economy settings. Fiscal dominance — the subordination of monetary policy to fiscal financing needs — has been identified as a primary driver of persistent inflation in several emerging market contexts. Exchange rate pass-through, whereby currency depreciations transmit rapidly into domestic price levels, further complicates the task of disinflation. Additionally, the formation of inflation expectations in environments with histories of high inflation tends to be backward-looking and adaptive, amplifying persistence.

While the literatures on inflation persistence and monetary policy effectiveness have developed largely in parallel, relatively few studies have examined these phenomena jointly within a unified empirical framework for emerging markets. This paper addresses that gap by incorporating both dimensions into a single panel estimation approach, allowing for an assessment of the degree to which monetary policy instruments moderate the structural persistence of

inflation.

3. Theoretical Framework

3.1 The New Keynesian Phillips Curve

The analytical foundation of this study draws on the New Keynesian Phillips Curve (NKPC), which characterizes inflation as a function of both forward-looking expectations and current real economic conditions. In its standard form, the NKPC posits that inflation is driven by the expected future price level and the output gap — the deviation of actual output from its potential level. The introduction of price rigidities and nominal contracting into this framework generates a role for lagged inflation, giving rise to the hybrid NKPC specification widely used in empirical work.

Within this framework, inflation persistence arises when price- and wage-setters rely heavily on past inflation outcomes in forming expectations, rather than on model-consistent forward-looking projections. When the backward-looking component dominates, shocks to the price level dissipate slowly, producing the sustained deviations from target that characterize high-persistence environments. This dynamic is particularly pronounced in emerging economies, where institutional credibility may be insufficient to anchor forward-looking expectations firmly to the policy target.

3.2 Rational and Adaptive Expectations

Rational expectations theory posits that economic agents form expectations by efficiently processing all available information, including knowledge of the policy rule and the central bank's commitment to its target. Under this assumption, a credible and transparent monetary authority can influence the inflation process primarily through its effect on expectations, potentially achieving disinflation at relatively low output cost. Empirical evidence suggests, however, that expectation formation in many emerging economies is better characterized by adaptive or backward-looking behavior, in which past inflation realizations dominate the formation of future inflation expectations. This adaptive dynamics sustain persistence and reduce the power of forward guidance as a monetary policy tool.

3.3 The Taylor Rule

The monetary policy reaction function in this study follows the Taylor Rule framework, which characterizes the setting of the nominal policy interest rate as a function of the deviation of inflation from its target and the output gap. The standard Taylor Rule implies that central banks respond to above-target inflation with interest rate increases sufficiently large to raise the real interest rate — the so-called Taylor principle — thereby cooling aggregate demand and moderating price pressures. In emerging market settings, adherence to a rule-based monetary framework is itself a determinant of credibility, and departures from such frameworks may reinforce adaptive expectations and sustain persistence.

4. Data and Methodology

4.1 Data

This study employs an unbalanced panel dataset comprising annual observations for a sample of emerging economies over a multi-year period. Data are sourced from the World Bank's World Development Indicators and the IMF's International Financial Statistics, supplemented by country-level institutional data from relevant cross-national databases. The primary variables of interest include the headline CPI inflation rate, the central bank policy interest rate, the nominal exchange rate, the output gap (estimated via the Hodrick-Prescott filter), and a measure of fiscal balance relative to GDP. Institutional quality indicators are included to capture variation in central bank independence and policy credibility across the sample.

4.2 Model Specification

Inflation persistence is first measured through a univariate autoregressive model of the form:

$$\pi_{it} = \alpha + \beta\pi_{it-1} + \varepsilon_{it}$$

where π_{it} denotes the inflation rate for country i in period t , π_{it-1} is the one-period lagged inflation rate, and ε_{it} is the error term. The coefficient β captures the degree of persistence: values close to unity indicate near-unit-root behavior, while lower values imply faster mean reversion.

To assess the role of monetary policy and structural factors, we estimate the following fixed-effects panel model:

$$\pi_{it} = \alpha + \beta\pi_{it-1} + \gamma MP_{it} + \delta X_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$$

where MP_{it} represents the monetary policy instrument (policy interest rate), X_{it} is a vector of control variables including the exchange rate, output gap, and fiscal balance, α_i captures country fixed effects, and λ_t captures time fixed effects. Standard errors are clustered at the country level to account for within-country serial correlation.

5. Empirical Results and Discussion

5.1 Inflation Persistence

The baseline autoregressive estimates confirm that inflation is highly persistent across the sample. The coefficient on lagged inflation (β) is estimated at approximately 0.72 and is statistically significant at the one percent level. This value indicates that approximately 72 percent of any inflationary shock persists into the following period, and implies that shocks take multiple years to fully dissipate under prevailing policy conditions. This finding is broadly consistent with the prior literature on inflation dynamics in emerging economies, which has documented substantially higher persistence relative to advanced economy benchmarks.

5.2 Monetary Policy Effectiveness

The policy interest rate enters the panel specification with a negative and statistically significant coefficient, confirming that contractionary monetary policy exerts downward pressure on inflation. However, the magnitude of this effect is moderate, suggesting that the standard interest rate channel operates with considerable attenuation in the emerging market context. This finding is consistent with the hypothesis that structural and institutional constraints weaken monetary transmission in these settings.

5.3 Role of Structural Factors

Exchange rate pass-through is positive and statistically significant, indicating that currency depreciations translate into higher domestic price levels — a finding with important implications for the conduct of monetary policy in open emerging economies. The output gap also enters with a positive and significant coefficient, consistent with standard demand-pull inflation dynamics. Fiscal variables display a weaker but positive relationship with inflation, suggesting that fiscal imbalances contribute to inflationary pressures, though this channel appears less dominant than monetary and exchange rate factors within this sample.

Taken together, these results underscore that inflation persistence in emerging economies is a structural feature that monetary policy alone cannot readily eliminate. While interest rate policy retains meaningful effectiveness, its impact is constrained by the exchange rate environment, the credibility of fiscal frameworks, and the degree to which expectations remain anchored to the inflation target. Durable reductions in persistence are therefore likely to require complementary reforms addressing institutional quality, fiscal discipline, and expectation management.

6. Conclusion

This paper has examined the relationship between inflation persistence and monetary policy effectiveness in a panel of emerging economies. The empirical results confirm that inflation persistence is high — with a lagged inflation coefficient of approximately 0.72 — and that monetary policy, while statistically effective, operates under binding structural constraints. Exchange rate pass-through, output gap dynamics, and fiscal imbalances all contribute independently to inflationary pressures, limiting the scope for interest rate policy to achieve rapid disinflation.

These findings carry several policy implications. First, central banks in emerging economies should not rely exclusively on interest rate adjustments to achieve price stability; complementary measures to strengthen institutional credibility, anchor expectations, and reduce exchange rate volatility are essential. Second, the coordination of monetary and fiscal policy is critical: fiscal dominance undermines the transmission of monetary policy and sustains inflation persistence. Third, the management of exchange rate dynamics — through reserve accumulation, macroprudential policy, or exchange rate flexibility — may be as important for inflation control as the setting of policy interest rates.

Future research should extend this analysis in several directions. The application of dynamic panel estimators, such as the Arellano-Bond GMM approach, would address potential endogeneity concerns in the estimation of the lagged inflation coefficient. The use of higher-frequency data would allow for a more precise identification of monetary policy transmission lags. Finally, country-level case studies could illuminate the heterogeneous institutional contexts that drive variation in persistence and policy effectiveness across the emerging market sample.

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